

session is available to meet demands coming from multiple locations.

It is to be noted that the programme's effectiveness, like any training, is always dependent on the training content design, duration of the training period and instructors' skills.

The future of e-learning: Enabling engagement, empowerment and evaluation (3-E)

With the success of CBT, WBT, LMS and virtual classrooms at the enterprise level, the next development in e-learning is point solutions that are specific to a segment of employees or enterprise. This can be done by using the increased computing power of devices, state-of-art technologies combined with application of Artificial Intelligence/Machine Learning and new learning methodologies first popularised by the Khan Academy and then followed by Coursera, Udemy, Udacity, Edx, etc.

Mobile and social technologies are two game changers for any enterprise with a mission to engage, empower and evaluate (the 3-E principle of learning) the new generation of learners. If one looks at the emerging trend in competency building, it is to reduce or eliminate the delay in:

- Providing training in what employees aspire to learn
- Discovering what the new/emerging technical skill demands are
- Finding out how employees learn by themselves

All of this is being done predominantly by transferring the responsibility for learning to employees, instead of the business/employer driving or owning the learning process — a turnaround of what has been in practice for the past two decades. The current and future adoption of e-learning tools and technologies is enabling this new mindset of the corporations as well as the new breed of learners, who are more than willing to learn by themselves in a well facilitated environment provided by the corporates.

The seeds for empowering learners were sown a decade ago, and have taken different forms as e-learning technologies have evolved.

- 1) The first seeds of empowerment were sowed when the e-learning material and books on programming were couriered to students placed in the IT company, but were in the final year of coursework in their colleges. This method gradually evolved to an online e-learning paradigm wherein students were auto registered to an e-learning course, post a job offer by the company that was accepted by the student.
- 2) The training environments are getting increasingly virtual with the advancement in software and hardware capacities and capabilities. The future is likely to see more innovation in the paradigm with added investment by IT corporations in campuses to enable e-learning through virtual environments that integrate campuses

(hardware and software) with the infrastructure of companies. One example could be e-learning courses blended with tele-presence technology.

- 3) To practice hands-on learning and for assessments, the training has been gradually moving to the cloud. This is a significant departure from the conventional process of learning and assessments, as not only the course material, classroom and operations but even the physical assets required for training are virtualised or digitised.
- The increased engagement levels of learners caused by creating learning experiences through digital technologies is promising for the IT companies.
- 4) Social technologies are at the centre stage of everything we do today. Purchase decisions are easily made using social ratings on products/services, and feedback on editorials and articles through social media, blogs, wikis, etc, is being used to collaboratively work on a project through crowd sourcing.
- 5) The devices that support virtual reality and augmented reality environments are getting affordable, as days pass. Adoption of these into learning models will aid in learning paradigms like DIY (do it yourself). It will also aid in the simplification of complex concepts through the creation of specialised e-learning content to be used in these environments, thus creating the next level of learning experiences that increase engagement, empowerment and evaluation.
- 6) New technologies like machine and deep learning will aid the implementation of more robust, adaptive e-learning materials.

Adaptive learning is unlike the conventional personalised learning, which is mostly based on preferences and the profile of learners. Adaptive learning changes the learning content in real-time as the machine learns the performance levels of the learner. This changes the course or lesson plans, further strengthening the 3-E principle of learning mentioned earlier. **END** 🐧

References

- [1] <https://github.com/dotnet/core>
- [2] <https://www.dotnetconf.net/>
- [3] <https://www.techempower.com/benchmarks/#section=data-r16&hw=ph&test=plaintext>

By: Ranganadhan N.S., Dr B. Thangaraju and Soji Joseph

Ranganadhan Nadadhur Srinivasa works at Wipro Technologies, Bengaluru. You can reach him at Ranganadhan.nadadhur@gmail.com.

Dr B. Thangaraju works at IIIT, Bengaluru. He can be reached at b.thangaraju@iiitb.ac.in.

Soji Joseph works at Wipro Technologies, Bengaluru. You can reach him at soji.joseph@wipro.com.